MIL-C-52828A(ME)
24 November 1981
SUPERSEDING
MIL-C-52828(ME)
23 July 1974

MILITARY SPECIFICATION

CARRIAGE, CABLEWAY

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the cableway carriage assembly which is a component of the Light Pioneer M-1 Aerial Cableway.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

FF-B-575	- Bolts, Hexagon and Square.
FF-N-836	- Nut: Square, Hexagon, Cap, Slotted, Castle, Knurled, Welding and Single Ball Seat.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5420

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MILITARY

MIL-T-704	- Treatment and Painting of Material.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.

STANDARDS

FEDERAL

FED-STD-H28	- Screw Thread Standards for Federal Services.
FED-STD-595	- Colors.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking and Waterproofing with Appropriate Test Methods.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.
MS15003	- Fittings, Lubrication (Hydraulic) Surface Check, 1/8 Pipe Threads, Steel, Type III.
MS15795	- Washer, Flat - Metal, Round, General Purpose (In./mm).
MS24466	- Bearing, Roller, Needle - Double Row, Heavy Duty, Track Roller, Type VI, Antifriction.
MS24665	- Pin, Cotter (Split).
MS35338	- Washer, Lock - Spring, Helical, Regular (Medium) Series, (In./mm).

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issue of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A108 - Steel Bars, Carbon, Cold Finished, Standard Quality.
A120 - Pipe, Steel, Black and Hot Dipped Zinc Coated (Galvanized) Welded and Seamless, for Ordinary Uses.
A668 - Steel Forgings Carbon & Alloy for General Industrial Use.
E18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

J435 - Automotive Steel Castings.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA. 15096.)

3. REQUIREMENTS

3.1 Description. The carriage shall be as shown in figures 1 and 2 and as specified herein.

3.2 Material. Material shall be as specified herein. Material not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.3).

3.2.1 Steel bar. Steel bar, including the steel used in the sheave shaft, shall conform to ASTM A108, UNS G10180 or G10200, size and form as applicable.

3.2.2 Cast steel. Cast steel for sheaves shall conform to SAE J435, grade 0050A.

3.2.2.1 Hardness. The cable groove of the trolley sheave shall be finished and either flame or induction hardened to a depth of 0.05 to 0.075 inch at a Rockwell hardness of C45 to C55.

3.2.3 Steel pipe. Steel pipe shall conform to ASTM A120 standard weight (Schedule 40), black plain ends, except that a hydrostatic test is not required.

3.2.4 Shackles. Shackles shall be made from steel conforming to ASTM A668, class F.

3.3 Components.

3.3.1 Bearings. Bearings shall conform to MS24466-10.

3.3.2 Nuts. Nuts shall conform to FF-N-836. The nuts for the trolley sheave shaft shall be type V, grade A. All other nuts shall be type II, style 4, grade A.

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3.3.2.1 Screw threads. Screw threads shall be in accordance with FED-STD-H28, 11 UNC, class 2A.

3.3.3 Flat washers. Flat washers shall conform to MS15795-820.

3.3.4 Bolts. Bolts shall conform to FF-B-575, type II, not less than grade 5, phosphate coated. Four 5/8 inch by 4-inch bolts and two 5/8 inch by 4-1/2 inch bolts are needed for complete assembly of the carriage. Not more than three bolt threads will extend beyond tightened nuts.

3.3.5 Lockwashers. Lockwashers shall conform to MS35338-145.

3.3.6 Cotter pins. Cotter pins shall conform to MS24665-374.

3.3.7 Lubrication fittings. Lubrication fittings shall conform to MS15003-1.

3.4 Treatment and painting. The portions of the carriage normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type A, except that the color shall conform to color number 14064 of FED-STD-595. Bolts, nuts, bearing surfaces, and threaded ends of components shall not be painted.

3.5 Workmanship.

3.5.1 Castings and forgings. All parts, components, and assemblies of the carriage which include castings and forgings shall be clean of harmful extraneous material such as sand, dirt, sprues, scale, and flux. Rework shall be limited to procedures which do not reduce strength or affect function. All rough edges shall be ground smooth.

3.5.2 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame cutting, using tips suitable for the thickness of the metal may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated metal shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges, and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.5.3 Machine work. Tolerances and gages for metal fits shall conform to the limits specified in figure 2 and to the standards of the industry. All burrs and rough edges shall be ground smooth.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for

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the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured and examined in accordance with referenced specifications and standards.

4.2 Classification of inspections. Inspections shall be classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of packing (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105, inspection level III.

4.3.2 Examination.

4.3.2.1 Samples. Samples selected in accordance with 4.3.1 shall be examined for the major and minor defects specified in 4.4.1. AQL shall be 0.4 percent defective for major defects and 4.0 percent defective for minor defects.

4.3.3 Tests.

4.3.3.1 Samples. Samples selected in accordance with 4.3.1 shall be tested as specified in 4.4.2.1. AQL shall be 4.0 percent defective.

4.4 Inspection procedure.

4.4.1 Examination. The carriage shall be examined as specified herein for the following defects:

Major

101. Carriage not as shown in figures 1 and 2.
102. Material not as specified.
103. Components not as specified.
104. Workmanship not as specified.

Minor

201. Treatment and painting not as specified.

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4.4.2 Tests.

4.4.2.1 Hardness. The cable groove of the trolley sheave shall be tested in accordance with ASTM E18. Nonconformance with 3.2.2.1 shall constitute failure of this test.

4.5 Inspection of packaging.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

105.	Bearings not lubricated as specified.
106.	Materials and containers not as specified for level A or B. Each incorrect material or container shall be considered one defect.
107.	Quantities packed together exceed the weight limitation of the box for level A or B.
108.	Carriages not blocked and braced within the boxes to prevent free movement and damage for level A or B.
109.	Strapping not as specified for level A.
110.	Packing not in accordance with the referenced document for commercial.
111.	Marking missing, illegible, incorrect or incomplete for level A, level B, or commercial.

5. PACKAGING

5.1 General requirement. The roller bearings shall be lubricated with grease conforming to MIL-G-23827.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Level A. The carriages shall be packed in close-fitting boxes conforming to PPP-B-601, overseas type, style optional; or PPP-B-621, class 2, style optional, in quantities not to exceed the weight limitations of the box. The contents shall be blocked and braced within the box in a manner to prevent free movement and damage in accordance with MIL-STD-1186. Box closure and strapping shall be in accordance with the appendix to the applicable box specification. Strapping shall conform to QQ-S-781, class 1, type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be finish B. When specified (see 6.2), strapping shall be finish A.

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5.2.2 Level B. The carriage shall be packed as specified in 5.2.1 except that boxes shall be domestic type or class as applicable. Strapping shall be in accordance with the appendix to the applicable box specification.

5.2.3 Commercial. The carriage shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection (level A or B) shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The carriage is intended for use as a traveling support for transporting either a carrier or a sling load over a cableway. NSN 5420-00-371-987 which is used in equipment set 5420-97-CL-E32 applies thereto.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Degree of packing required (see 5.2).
(c) When other than finish B strapping is required (see 5.2.1).

6.3 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.2).

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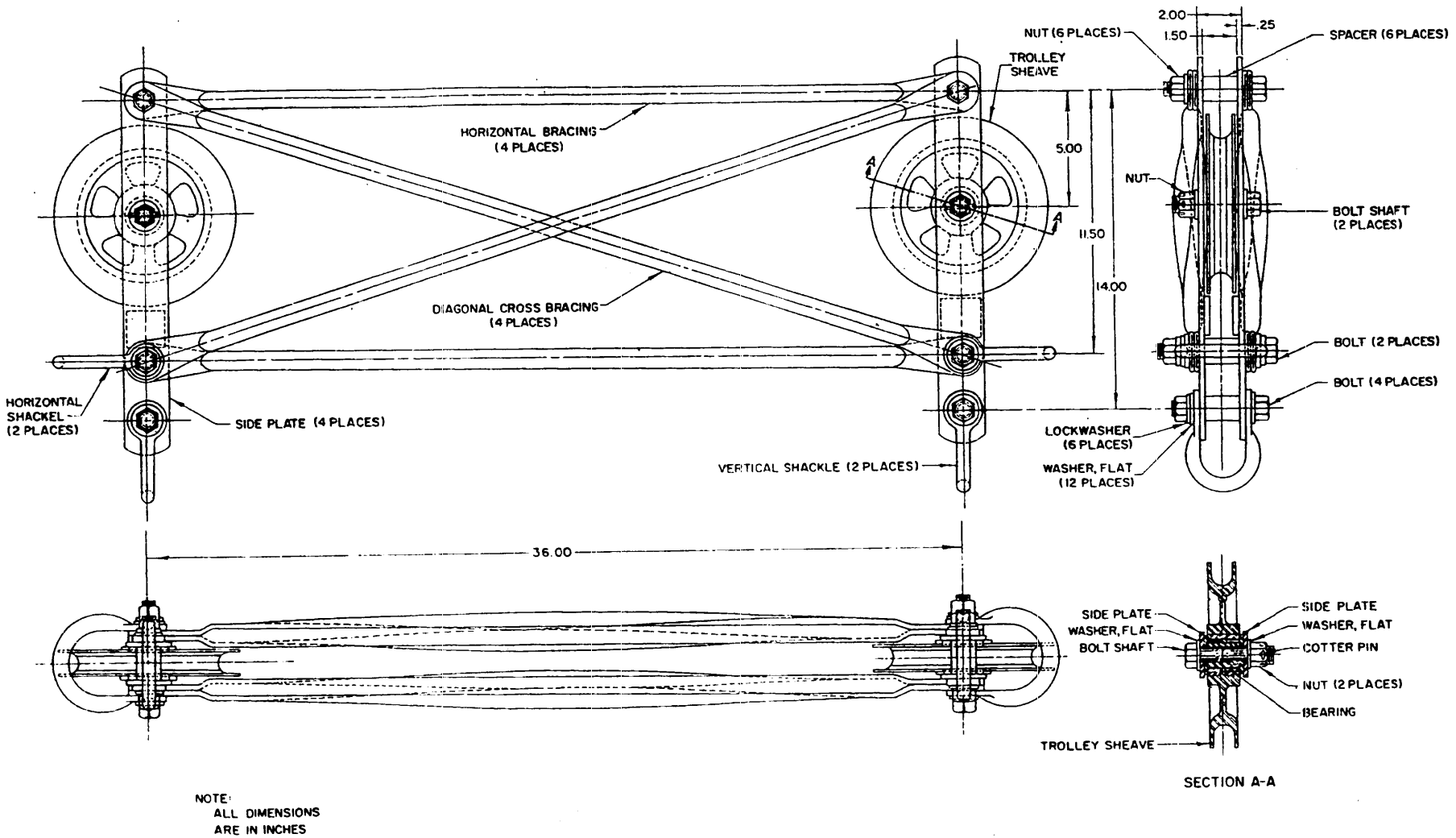
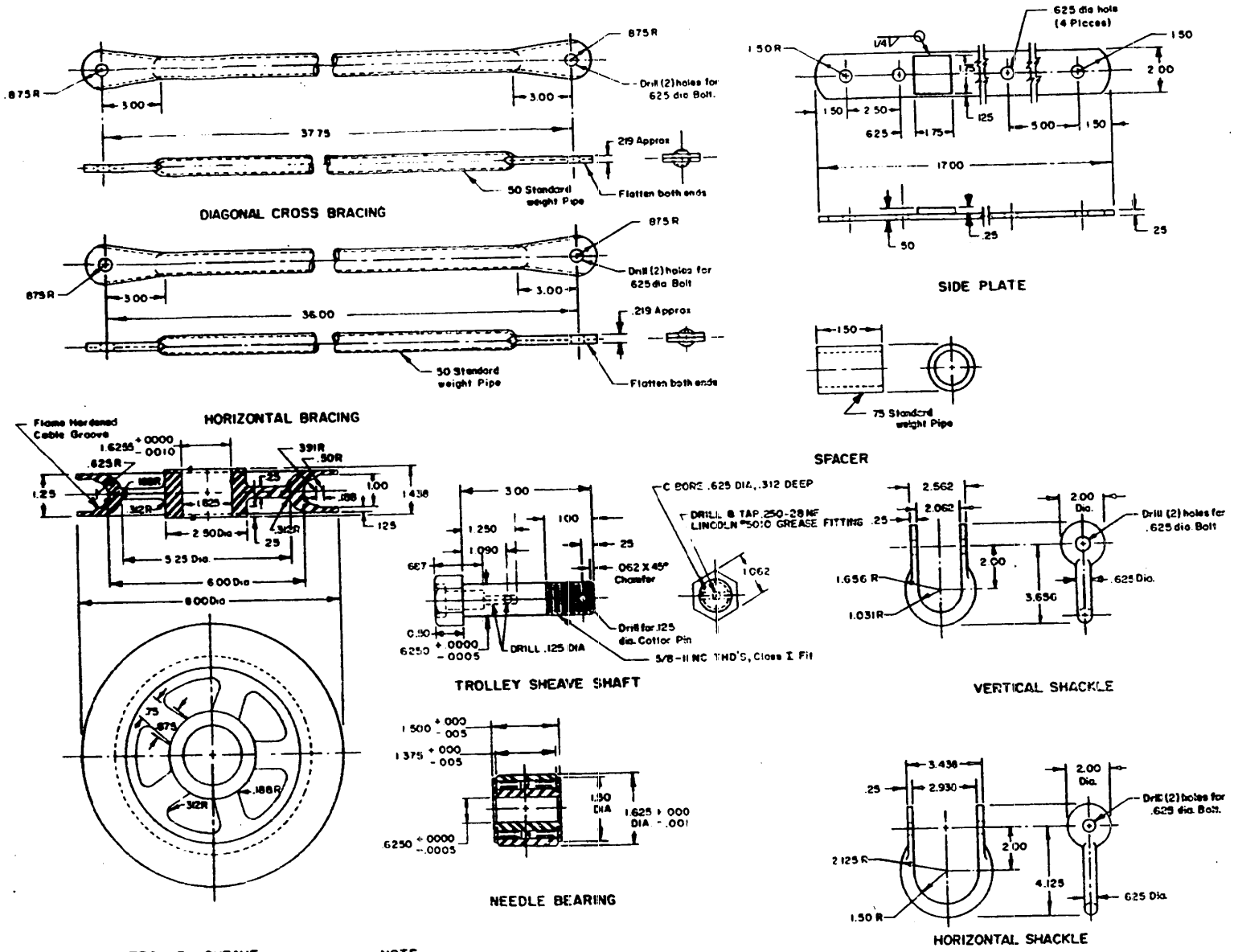


FIGURE 1. Cableway carriage, general assembly.

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NOTE:
SHAFT TO BE CYANIDE HARDENED, DEPTH OF CASE 0.005-0.010 IN., 89 HR15N MINIMUM.
ALL DIMENSIONS IN INCHES

FIGURE 2. Cableway carriage components.

X-27318

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